Fertility and retirement policies

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Relaxing fertility policies and delaying retirement age increase China's carbon emissions

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Relaxing fertility policies and delaying retirement age would increase China's household carbon footprint mainly by boosting population and labour. Policymakers should synergize policies targeting population ageing and climate change, which are both crucial for sustainable development.

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The policy problem

The world is witnessing a demographic shift towards ageing societies, posing a serious challenge to sustainable development and climate actions. In response, China has progressively relaxed its fertility policy (that is, shifting from the one-child policy to the two-child policy, and now to the three-child policy) and has proposed a gradual increase in the statutory retirement age (currently 60 for men and 55 for women, compared with 65 for both men and women in developed countries). These policy changes are expected to impact population size, age structure and the labour market, potentially influencing household consumption and carbon footprint. However, such impacts lack quantitative assessment. Capturing the potential impacts of relaxing fertility policies and delaying retirement age on household carbon footprints is critical for policymakers to coordinate demographic policies with those for climate change mitigation.

The findings

We find that younger people in China tend to have higher household carbon footprints due to greater income and consumption, a pattern that contrasts with developed countries where older people typically have higher carbon footprints. Relaxing fertility policies and delaying retirement age are projected to increase household carbon footprints in China, primarily through boosting the population and labour. These impacts of the policy changes seem to be greater in regions where disparities in income and consumption among different age groups are larger. These findings do not imply that such policies targeting population ageing should be avoided to alleviate environmental pressure. Instead, they provide evidence of the interactions between policies targeting population ageing and climate change, and highlight the importance of synergizing policies with different targets.

The study

We developed an integrated framework of climate, population and economy models to assess the impacts of fertility and retirement policies on household carbon footprints. We estimate household carbon footprints in different age groups by combining the environmentally extended multiregional input–output model with detailed household survey data. Fertility and retirement policies affect household carbon footprints via two channels. First, they affect the population in terms of size and structure, which is assessed by using the cohort-component method. Second, they also affect the supply of the labour force and subsequently the economy, which is estimated by analysing changes in labour force participation rates and household consumption patterns across age groups.

Recommendations for policy

- Assessing the climate impact is crucial when developing fertility and retirement policies.
- Promoting greener consumption and sustainable lifestyles, particularly among young people, is an effective measure to reduce household carbon footprints in China.
- Reducing income and consumption disparities between age groups could alleviate the impacts of fertility and retirement policies on household carbon footprints.
- Delaying retirement can considerably lower dependency ratios and enhance demographic dividends, although it may increase household carbon footprints.
- Policymakers need to consider the synergy between policies addressing population ageing and those addressing climate change.

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Further reading

1. Mi, Z. et al. Economic development and converging household carbon footprints in China. *Nat. Sustain.* **3**, 529–537 (2020).
This paper provides an environmentally extended multiregional input–output analysis to estimate household carbon footprints in China.
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This literature review provides a systematic overview of China's demographic change and its impacts.
3. Yu, B., Wei, Y., Kei, G. & Matsuoka, Y. Future scenarios for energy consumption and carbon emissions due to demographic transitions in Chinese households. *Nat. Energy* **3**, 109–118 (2018).
This paper analyses the impacts of demographic transitions on energy consumption and carbon emissions in Chinese households.
4. Zeng, Y. & Hesketh, T. The effects of China's universal two-child policy. *Lancet* **388**, 1930–1938 (2016).
This literature review documents evidence for the potential effects of China's universal two-child policy on demographic, health, social and environmental aspects.
5. Zheng, H. et al. Ageing society in developed countries challenges carbon mitigation. *Nat. Clim. Change* **12**, 241–248 (2022).
This paper shows that older people in developed countries play a leading role in driving up greenhouse gas emissions due to their carbon-intensive lifestyle.

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Competing interests
The authors declare no competing interests.